Integration by Parts

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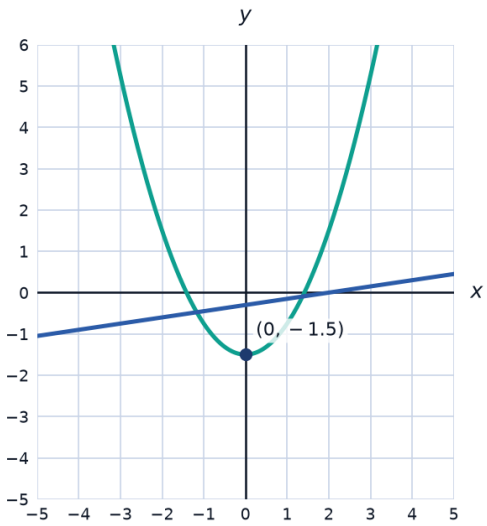
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The integration by parts formula

State the integration by parts formula, and explain the LIATE guideline for choosing u.

2 [Foundational] • Target: The integration by parts formula

Derive the integration by parts formula starting from the product rule $\frac{d}{dx}[uv] = u'v + uv'$.



3 [Foundational] • Target: Repeated integration by parts

Evaluate $\int x^2 e^x \, dx$, applying integration by parts twice.

4 [Foundational] • Target: Repeated integration by parts

Set up (do not evaluate fully) the reduction needed to find $\int x^3 e^x \, dx$, identifying how many times integration by parts must be applied.

5 [Foundational] • Target: The cyclic case

Evaluate $\int e^x \sin x \, dx$ using the cyclic (solve-for-the-integral) technique.

6 [Foundational] • Target: The cyclic case

Evaluate $\int e^{2x} \cos x \, dx$ using the cyclic technique.

7 [Foundational] • Target: Definite integrals using integration by parts

Evaluate $\int_0^1 x e^x \, dx$ using the antiderivative found earlier.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Integration by parts to derive a reduction formula

Use integration by parts to show $\int x^n e^x\,dx = x^n e^x - n\int x^{n-1}e^x\,dx$ (the reduction formula), and use it once to reconfirm $\int x^2 e^x\,dx$.

9 [Intermediate] • Target: Choosing u and dv wisely

For each integral, state the best choice of u (using LIATE) without evaluating:

a $\int x^3 \ln x\,dx$

b $\int xe^{5x}\,dx$

c $\int x\tan^{-1}x\,dx$

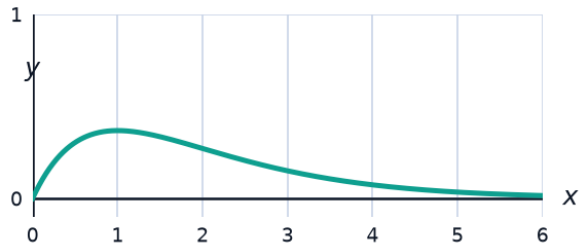
d $\int x^2\cos x\,dx$

10 [Intermediate] • Target: Choosing u and dv wisely

Explain what goes wrong if you attempt $\int xe^x\,dx$ with the reversed choice $u = e^x$, $dv = x\,dx$.

11 [Intermediate] • Target: Definite integrals: area under a transcendental product

Find the exact area under $y = xe^{-x}$ on $[0, 3]$, using $\int xe^{-x}\,dx = -xe^{-x} - e^{-x} + C$ (verify this antiderivative first by differentiating).



12 [Intermediate] • Target: Mixed review: choosing the right technique

For each integral, state whether it should be solved by u-substitution or integration by parts, and identify the key choice (substitution variable or u) without fully evaluating:

a $\int 2xe^{x^2}\,dx$

b $\int x^2 e^x\,dx$

c $\int \frac{\ln x}{x}\,dx$

d $\int x\ln x\,dx$

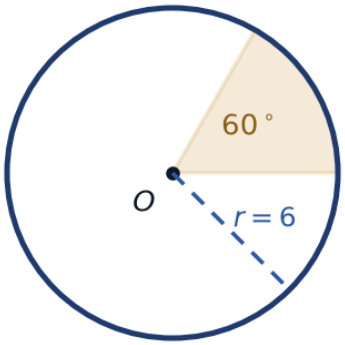
13 [Intermediate] • Target: Repeated integration by parts

Level 3 · Advanced / AP FRQWrite the definite integral that represents the total perimeter (arc length) of the cardioid $r = 1 + \cos\theta$, and simplify the integrand algebraically into a single trigonometric function.

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14 **[Intermediate]** • **Target: The cyclic case**

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2 \cos x \, dx$.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Definite integrals using integration by parts** • **Cross-Lesson: 4.01 Polar Curves and Derivatives**

Level 2 · MediumEvaluate $\int x^3 \sin(2x)\, dx$ using the Tabular Method.

16 **[Challenging]** • **Target: Integration by parts to derive a reduction formula** • **Cross-Lesson: 4.02 Area in Polar Coordinates**

Level 3 · Advanced / AP FRQA particle moves along a planar path with velocity $\vec{v}(t) = \langle \sin(t^2), \cos(t^3) \rangle$ for $0 \le t \le 3$. Write the definite integral representing the total distance traveled by the particle, and calculate the numerical value using three decimal places.

17 **[Challenging]** • **Target: Choosing u and dv wisely** • **Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests**

[Unit 6BC · Advanced Integration Techniques · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 **[Challenging]** • **Target: Integration by parts with algebraic-trigonometric products** • **Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests**

[Unit 9 · Polar Coordinates & Polar Area · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 **[Challenging]** • **Target: Definite integrals: area under a transcendental product** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x \, dx$ for integers $n \ge 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \ge 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x \, dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n \to \infty} I_n$.

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20 [Challenging] • Target: Mixed review: choosing the right technique • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Level 3 · AdvancedLet $h(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. Prove that h is strictly increasing on $\mathbb{R}$ without using calculus derivatives, deduce that h is injective, and derive the explicit formula for $h^{-1}(x)$ including its domain.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The integration by parts formula

Evaluate $\int_0^{\pi} -2x\cos x\,dx$, using integration by parts.

a. 2

b. 6

c. 4

d. -4

22 [Foundational] • Target: Integration by parts with logarithms and inverse trig

Given $f(0) = 0$, $f(3) = 2$, $g(0) = 2$, $g(3) = 2$, and $\int_0^3 f'(x)g(x)\,dx = -2$, use integration by parts to find $\int_0^3 f(x)g'(x)\,dx$.

a. 10

b. -6

c. 2

d. 6

23 [Foundational] • Target: Repeated integration by parts

Using the cyclic technique, $\int e^x\cos(2x)\,dx = \frac{e^x(\cos(2x) + C_1\sin(2x))}{5} + C$. Find the coefficient C_1.

a. $\frac{1}{5}$

b. $-\frac{2}{5}$

c. 2

d. $\frac{2}{5}$

24 [Foundational] • Target: The cyclic case

For $\int x\ln x\,dx$, which is the correct choice of u according to LIATE?

a. It does not matter which factor is chosen for u

b. $u = x$ (Algebraic always goes first)

c. $u = x$ (the opposite LIATE priority)

d. $u = \ln x$ (Logarithmic outranks Algebraic in LIATE)

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25 [Foundational] • Target: Definite integrals using integration by parts

Consider a region R in the first quadrant bounded by the curve $y = e^{-x^2}$, the x-axis, and the y-axis. Let S be the solid generated when R is revolved about the y-axis. The volume of S can be calculated using the method of cylindrical shells or the disk/washer method. However, direct application of standard integration techniques to find the volume of S leads to an integral that cannot be evaluated in terms of elementary functions. A related problem involves finding the surface area of a solid of revolution. Let C be the curve defined by $y = \sqrt{1 - x^2}$ for $0 \le x \le 1$. This curve represents a quarter-circle. If this curve C is revolved about the x-axis, it generates a hemisphere. The surface area of this hemisphere (excluding the base) is given by the integral $A = \int_0^1 2\pi y\sqrt{1 + (y')^2}\,dx$. Determine the exact surface area of the generated hemisphere.

a. $\frac{\pi}{2}$

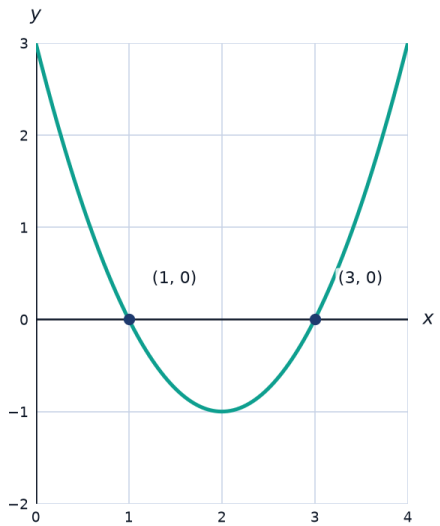
b. π

c. 2π

d. 4π

26 [Foundational] • Target: Integration by parts to derive a reduction formula

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. What is the definite integral of $f(x)$ from $x = 1$ to $x = 3$?



a. $-\frac{4}{3}$

b. 0

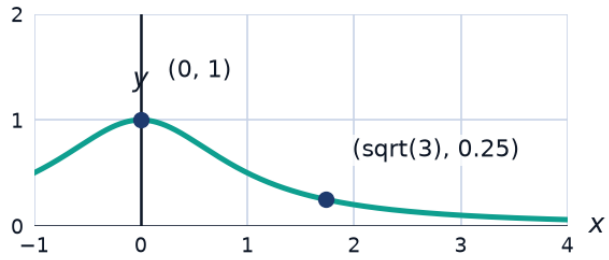
c. $\frac{4}{3}$

d. 2

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27 [Foundational] • Target: Choosing u and dv wisely

Consider the region bounded by the curve $y = \frac{1}{x^2 + 1}$, the x-axis, and the vertical lines $x = 0$ and $x = b$, where $b > 0$. If the area of this region is $\frac{\pi}{3}$, what is the value of b?



a. $\sqrt{2}$

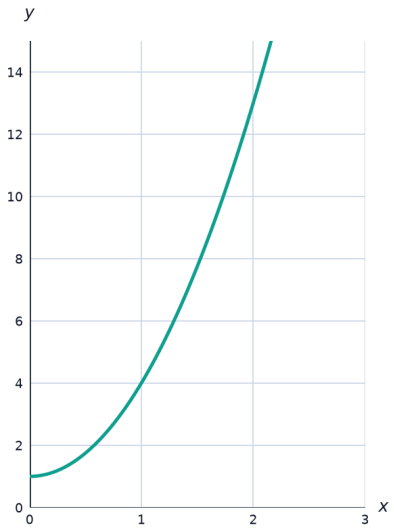
b. $\frac{1}{\sqrt{3}}$

c. $\sqrt{3}$

d. 2

28 [Foundational] • Target: Integration by parts with algebraic-trigonometric products

A region in the first quadrant is bounded by the graph of $f(x) = 3x^2 + 1$, the x-axis, and the vertical lines $x = 1$ and $x = 2$. What is the area of this region?



a. 10

b. 12

c. 14

d. 16

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Definite integrals: area under a transcendental product

Using the reduction formula, evaluate $\int_0^{\frac{\pi}{2}} \cos^6(x)\,dx$.

a. $\frac{5\pi}{16}$

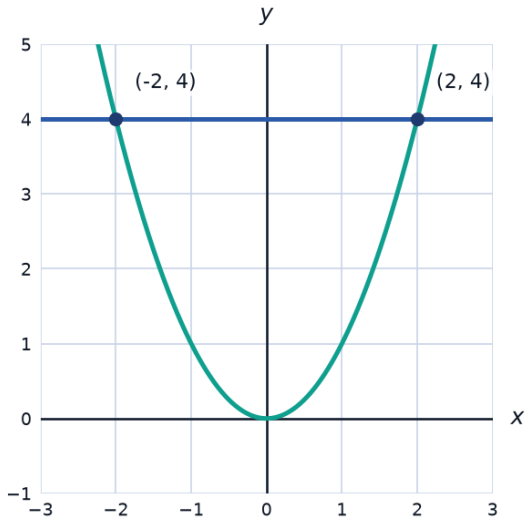
b. $\frac{5\pi}{32}$

c. $\frac{3\pi}{16}$

d. $\frac{1}{6}$

30 [Intermediate] • Target: Mixed review: choosing the right technique

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?



a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

31 [Intermediate] • Target: The integration by parts formula

Consider the function represented by the graph of $f(x)$ below. What is the value of the definite integral $\int_0^4 f(x)\,dx$?

a. 12

b. 10

c. 8

d. 6

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32 [Intermediate] • Target: Integration by parts with logarithms and inverse trig

Evaluate $\int x^2 e^{2x}\,dx$ (apply integration by parts twice).

a. $\left(\frac{x^2}{2} + \frac{2x}{4} + \frac{2}{8}\right)e^{2x} + C$

b. $\left(\frac{x^2}{2} - \frac{2x}{4} + \frac{2}{8}\right)e^{2x} + C$

c. $\left(\frac{x^2}{2} - \frac{2x}{4}\right)e^{2x} + C$

d. $x^2e^{2x} - 2xe^{2x} + 2e^{2x} + C$

33 [Intermediate] • Target: Repeated integration by parts

Integration by parts: $\int u\,dv =$

a. $uv + \int v\,du$

b. $uv - \int v\,du$

c. $u \cdot v$

d. $\frac{u}{v}$

34 [Intermediate] • Target: The cyclic case

For $\frac{1}{(x-2)^2(x-6)} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{D}{x-6}$, find B (coefficient of the repeated-factor term).

a. $-\frac{1}{2}$

b. $-\frac{1}{4}$

c. $-\frac{3}{20}$

d. $\frac{1}{4}$

35 [Intermediate] • Target: Definite integrals using integration by parts

Convert the polar point $(4,\ \pi/4)$ to rectangular coordinates.

a. $(2\sqrt{2},\ 2\sqrt{2} + 1)$

b. $(2\sqrt{2},\ 2\sqrt{2})$

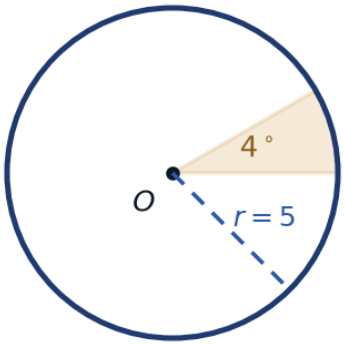
c. $(-2\sqrt{2},\ 2\sqrt{2})$

d. $(4,\ \pi/4)$

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36 **[Intermediate]** • **Target: Integration by parts to derive a reduction formula**

A point is represented in polar coordinates by $(r, \theta) = (5, \frac{\pi}{4})$. What are the equivalent Cartesian coordinates (x, y)?



a. $(\frac{5\sqrt{2}}{2}, \frac{5\sqrt{2}}{2})$

b. $(5, 5)$

c. $(\frac{5}{2}, \frac{5}{2})$

d. $(\frac{5}{2}, \frac{5\sqrt{3}}{2})$

37 **[Intermediate]** • **Target: Choosing u and dv wisely**

$\int \sin^4 x \, dx$ using reduction formula:

a. $\frac{3x}{4} + \frac{\sin 2x}{4} + C$

b. $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

c. $\frac{x}{2} - \frac{\sin 4x}{8} + C$

d. $\frac{\sin^3 x \cos x}{4} + \frac{\sin x \cos x}{4} + \frac{x}{8} + C$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Integration by parts with algebraic-trigonometric products • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Consider the region R in the first quadrant bounded by the curve $y = e^{-x^2}$, the x-axis, and the line $x = a$, where $a > 0$. A solid is generated by revolving this region R about the y-axis. If the volume of this solid is $\frac{\pi}{2}$, determine the value of a.

a. $\ln(2)$

b. $\sqrt{\ln(2)}$

c. $e^{\sqrt{2}}$

d. $\sqrt{2}$

39 [Challenging] • Target: Definite integrals: area under a transcendental product • Cross-Lesson: 6.01 Power Series and Interval of Convergence

A plane region R is bounded by the graph of $y = e^{-x^2}$, the x-axis, and the lines $x = 0$ and $x = 1$. A solid is generated by revolving the region R about the line $y = -2$. The cross-sections of the solid perpendicular to the x-axis are squares. The volume of this solid is given by $V = \int_a^b A(x)dx$. (a) Find the outer radius and inner radius of a representative disk if the region were revolved about the line $y = -2$ using the washer method. (b) Set up, but do not evaluate, an integral expression for the volume of the solid generated by revolving the region R about the line $y = -2$ using the washer method. (c) Set up, but do not evaluate, an integral expression for the volume of the solid generated by taking cross-sections perpendicular to the x-axis that are squares. (d) Explain why the integral expression in part (c) is the correct representation for the volume of the solid.

a. The radius of a disk in the washer method is $R(x) = e^{-x^2} + 2$ and $r(x) = 2$.

b. The volume using the washer method is $V = \int_0^1 \pi((e^{-x^2} + 2)^2 - 2^2)dx$.

c. The area of a square cross-section is $A(x) = (e^{-x^2})^2$.

d. The integral expression in part (c) represents the volume because it sums the areas of square cross-sections perpendicular to the x-axis over the interval [0, 1].

40 [Challenging] • Target: Mixed review: choosing the right technique • Cross-Lesson: 6.02 Taylor and Maclaurin Series

A cardioid is described by the polar equation $r = a + b\sin(\theta)$, where a and b are positive constants. The graph of this cardioid passes through the point $(r, \theta) = (5, \frac{\pi}{2})$ and the point $(r, \theta) = (3, \frac{3\pi}{2})$. What is the value of the parameter a?

a. 2

b. 3

c. 4

d. 5

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41 [Challenging] • Target: The integration by parts formula • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

The rate of change of a quantity $Q(t)$ is given by $\frac{dQ}{dt} = kte^{-0.5t^2}$, where k is a positive constant. If $Q(0) = 5$ and $Q(2) = 15$, find the value of k.

a. $\frac{5}{1-e^{-2}}$

b. $\frac{5}{e^{-2}-1}$

c. $\frac{10}{1-e^{-2}}$

d. $\frac{10}{e^{-2}-1}$

42 [Challenging] • Target: Integration by parts with logarithms and inverse trig • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

What is the area of the region bounded by the polar curve $r = \sqrt{5\sin\theta}$ for $0 \le \theta \le \pi$?

a. 5

b. 10

c. $\frac{5}{2}$

d. 5π

43 [Challenging] • Target: Repeated integration by parts • Cross-Lesson: 3.01 Parametric Equations

The graph of the function $f(x)$ is shown below. What is the value of the definite integral $\int_{-2}^{2} f(x)\,dx$?

a. 0

b. 2

c. 4

d. 8

44 [Challenging] • Target: The cyclic case • Cross-Lesson: 3.02 Vector-Valued Functions

A population of bacteria follows a logistic growth model where $P(t)$ is the population at time t (in hours). The differential equation governing the population growth is $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{1000}\right)$. If the initial population is $P(0) = 200$, use Euler's method with a step size of $\Delta t = 1$ hour to approximate the population after 1 hour, $P(1)$.

a. 210

b. 208

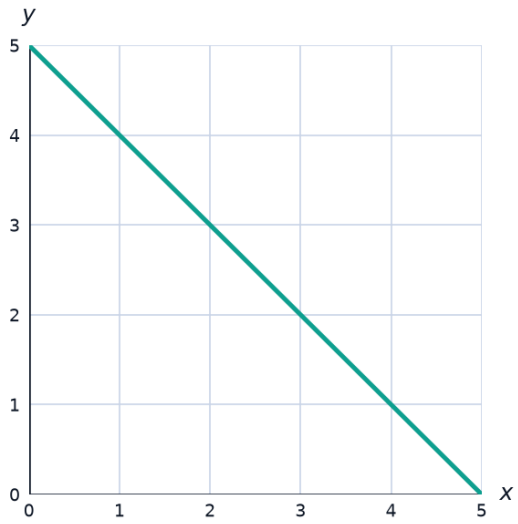
c. 212

d. 209

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45 [Challenging] • Target: Definite integrals using integration by parts • Cross-Lesson: 4.01 Polar Curves and Derivatives

A differentiable function $f(x)$ is represented by the graph below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 1$ to $x = 4$ is given by the definite integral $\int_1^4 f(x)\,dx$. If $f(x) = -x + 5$ for $1 \le x \le 4$, what is the area of this region?



a. 4.5

b. 6

c. 7.5

d. 9

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